



Model Types and the Simulation Study

What You Should Be Able to Do

- Classify any model by time, randomness, and state change.
- Write one complete classification sentence with reasons.
- Explain how the classification changes the experiment.
- Outline a study from the decision question to usable evidence.

Classification Prevents Wrong Models

- **Time:** tells us whether a simulated clock and run length are needed.
- **Randomness:** tells us whether repeated runs and statistics are needed.
- **State change:** tells us whether to use equations, events, or both.
- **Main idea:** classification is a design decision, not just vocabulary.

Ask These Three Questions in Order

- 1. Time:** Does behavior unfold over simulated time?
 - 2. Randomness:** Can the same situation produce different outcomes?
 - 3. State change:** Does the state move smoothly or jump at events?
- Answer each question independently before writing the final classification.

The Question Controls the Model

The real system is a bank, but the model depends on the decision question.

- **Question A:** What is the total balance today? A snapshot may be static.
- **Question B:** How long will customers wait at lunch? Time must be represented.
- **Takeaway:** classify the model being built, not the name of the real system.

Question 1: Does Simulated Time Matter?

1. **Ask:** Can the timing or order of changes affect the answer?
 2. **If no:** the model is static.
 3. **If yes:** the model is dynamic.
- **Important:** computer processing time is not the same as simulated system time.

Static Model: No Timeline Is Needed

- **Meaning:** the model represents a snapshot, final result, or atemporal experiment.
- **What is missing:** it does not track state from one simulated time to another.
- **Example:** estimate π using randomly placed points.
- **Why static:** the samples do not represent a system clock.

Dynamic Model: Order and Timing Matter

- **Meaning:** the model tracks state or performance as simulated time advances.
- **Example:** customers arrive, wait, receive service, and leave.
- **Why dynamic:** changing arrival order or timing can change the waiting time.
- **Possible outputs:** queue length over time, delay, utilization, or inventory level.

One System Can Use Either Time Model

- **Static bank question:** What is the total balance at this moment?
- **Dynamic bank question:** How does the queue change from 12:00 to 1:00 PM?
- **Difference:** the second question requires a timeline and event order.
- **Decision rule:** choose dynamic only when time can change the answer.

Quick Check: Static or Dynamic?

- **Case A:** estimate the probability that a component is defective.
- **Answer A:** static, because no simulated timeline is required.
- **Case B:** track machine failures and repairs during one week.
- **Answer B:** dynamic, because failures and repairs occur over time.

Question 2: Can the Same Run Change?

1. **Imagine:** run the same scenario again with the same starting conditions.
 2. **Same output:** the model is deterministic.
 3. **Different possible output:** the model is stochastic.
- **Look for:** random inputs or random behavior inside the model.

Deterministic: Same Inputs, Same Output

- **Meaning:** the model contains no random variables.
- **Rule:** the same inputs and starting state always produce the same result.
- **Example:** chemical concentration follows a known decay equation.
- **Analysis:** compare scenarios or test how results change when inputs change.

Stochastic: Random Inputs Change Results

- **Meaning:** the model contains one or more random variables.
- **Rule:** the same scenario can produce different outputs across runs.
- **Example:** customer arrival and service times vary.
- **Analysis:** use multiple replications and summarize the distribution of results.

Same Queue, Same Means, Different Behavior

- **Deterministic:** one customer arrives every minute; service always takes 0.99 minute.
- **Stochastic:** interarrival and service times vary around the same averages.
- **Same means:** both models use 1.00 and 0.99 minutes.
- **Different result:** only the stochastic model creates congestion from variability.

Queue Example: Read the Given Values

- **Mean interarrival time:** 1.00 minute between customer arrivals.
- **Mean service time:** 0.99 minute to serve one customer.
- **One server:** only one cashier processes the customers.
- **First impression:** service is slightly faster than arrivals, so the system looks safe.

Step 1: Convert Times into Rates

- **Arrival rate:** λ (lambda) = $1 / 1.00 = 1$ customer per minute.
- **Service rate:** μ (mu) = $1 / 0.99 = 1.0101$ customers per minute.
- **Meaning:** the cashier can serve only slightly more than one customer per minute.
- **Capacity gap:** $1.0101 - 1 = 0.0101$ customer per minute.

Step 2: Calculate Utilization

Utilization formula: ρ (rho) = arrival rate / service rate

Substitute: $\rho = 1 / 1.0101 = 0.99$

- **As a percentage:** the cashier is busy about 99% of the time.
- **Meaning:** only about 1% spare capacity remains to recover from delays.

Step 3: Calculate the M/M/1 Queue Delay

M/M/1 formula: $W_q = \rho / (\mu - \lambda)$

Substitute: $W_q = 0.99 / (1.0101 - 1)$

Result: W_q is approximately 98.01 minutes

- **Meaning:** this is the average waiting time before service begins.
- **Remember:** it is a steady-state average, not every customer's exact wait.

Why the Deterministic Queue Has No Wait

- **Customer 1:** arrives at 0.00 and finishes at 0.99.
- **Customer 2:** arrives at 1.00 and finishes at 1.99.
- **Customer 3:** arrives at 2.00 and finishes at 2.99.
- **Result:** each service ends 0.01 minute before the next arrival.
- **Therefore:** no customer needs to wait.

Main Lesson: Variability Needs Spare Capacity

- **Same averages:** both queue models use 1.00 and 0.99 minutes.
- **Deterministic result:** regular timing produces no queue.
- **Stochastic result:** random timing produces an average delay near 98 minutes.
- **Takeaway:** near 100% utilization, small variations can create severe congestion.

Quick Check: Deterministic or Stochastic?

- **Chemical decay:** known rate and starting value means deterministic.
- **Customer arrivals:** random interarrival times means stochastic.
- **Fixed random seed:** makes a stochastic experiment reproducible.
- **Important:** reproducibility does not remove randomness from the model.

Question 3: How Does the State Change?

- 1. Name the state:** identify the variable describing the system condition.
- 2. Smooth movement:** classify the state as continuous.
- 3. Jumps at events:** classify the state as discrete.
- **Both mechanisms:** classify the model as mixed or hybrid.

Continuous: State Moves Smoothly

- **Meaning:** the state can change at every instant.
- **Examples:** temperature, speed, pressure, and liquid level.
- **Typical logic:** equations describe the rate of change.
- **Example reason:** tank level is continuous because it can take any value in a range.

Discrete: State Jumps at Events

- **Meaning:** the state changes only when an event occurs.
- **Examples:** arrival, departure, failure, and repair.
- **Between events:** the discrete state remains unchanged.
- **Example reason:** number in queue jumps from 2 to 3 when a customer arrives.

Mixed: Both Kinds of Change Matter

- **Discrete part:** a factory has arrivals, failures, repairs, and job completions.
- **Continuous part:** the same factory tracks temperature or pressure.
- **Classification:** the model is mixed because it needs events and equations.
- **Modelling rule:** include both only when both can affect the decision.

Quick Check: Name the State First

- **Tank level:** continuous because the level changes smoothly.
- **Number in queue:** discrete because it changes at arrivals and departures.
- **Factory status and temperature:** mixed because one jumps and one moves smoothly.
- **Answer pattern:** name the state, classify its change, then give the reason.

Write One Complete Classification Sentence

Sentence frame:

The model is [static/dynamic], [deterministic/stochastic], and [continuous/discrete/mixed].

Example:

The queue model is dynamic because time affects waiting, stochastic because times vary, and discrete because state changes at events.

Worked Examples: Bank and Chemical Decay

- **Bank queue:** dynamic because waiting unfolds over time.
- **Bank queue:** stochastic because times vary; discrete because counts change at events.
- **Chemical decay:** dynamic because concentration changes over time.
- **Chemical decay:** deterministic because the equation is fixed; continuous because concentration changes smoothly.

Worked Examples: Factory and Lottery

- **Factory:** dynamic because operations unfold over time.
- **Factory:** stochastic because failures vary; mixed because status jumps while temperature changes smoothly.
- **Lottery outcome:** static because only the final draw result is represented.
- **Lottery outcome:** stochastic because the result is random; discrete because outcomes are distinct.

Classification Changes the Experiment

- **Stochastic:** use seeds, replications, distributions, and confidence intervals.
- **Deterministic:** compare scenarios and test parameter sensitivity.
- **Discrete:** implement event or process logic.
- **Continuous:** implement equations and numerical integration.
- **Dynamic:** choose a run length and possibly a warm-up period.

Choose the Simplest Credible Model

1. Start with the decision question, not the software.
 2. Include a behavior only if it could change the decision.
 3. Use a deterministic approximation when variability is unimportant.
 4. Use a static model when timing and event order do not matter.
- **Limit:** simple assumptions must still be credible for the intended decision.

Ten Steps Fit into Three Phases

- A. Define:** Steps 1-3 establish the question, data, model, and assumptions.
 - B. Build and check:** Steps 4-6 create a credible programmed model.
 - C. Experiment and use:** Steps 7-10 turn runs into decision evidence.
- **Remember:** new evidence may send the team back to an earlier phase.

Steps 1-3: Define Before You Build

- 1. Formulate:** state the decision question, objectives, measures, scope, and constraints.
 - 2. Collect and model:** gather data and describe the system logic conceptually.
 - 3. Validate assumptions:** review the conceptual model with stakeholders.
- **Output:** an agreed study plan and conceptual model.

Steps 4-6: Build and Check Credibility

4. Construct and verify: program the model and check that the logic was implemented correctly.

5. Make pilot runs: find input, logic, and output problems early.

6. Validate behavior: compare the model with data and expert expectations.

• **Key difference:** verification checks the build; validation checks the representation.

Steps 7-10: Produce Useful Evidence

- 7. Design experiments:** choose scenarios, run length, warm-up, and replications.
- 8. Make production runs:** execute the approved experiment plan.
- 9. Analyze output:** estimate performance and compare alternatives.
- 10. Document and use:** report assumptions, methods, limitations, and decisions.

Worked Study: Classify a Grocery Checkout

- **Decision question:** Will a second cashier reduce unacceptable waiting?
- **Dynamic:** waiting depends on arrival and service times.
- **Stochastic:** interarrival and service times vary.
- **Discrete:** queue length changes when customers arrive or depart.
- **Complete answer:** the checkout model is dynamic, stochastic, and discrete.

Worked Study: Plan the Grocery Experiment

- 1. Collect inputs:** measure interarrival and service times.
 - 2. Choose measures:** mean wait, 90th-percentile wait, and cashier utilization.
 - 3. Compare scenarios:** test one cashier versus two cashiers.
 - 4. Repeat runs:** use multiple replications because the model is stochastic.
- **Decision:** use the distribution of results, not one run.

Your Turn: Classify and Plan

Choose a grocery checkout, airport runway, factory, or computer network.

1. Write one decision question and one performance measure.
2. Answer the time, randomness, and state-change questions.
3. Write the complete classification sentence with reasons.
4. Outline Steps 1-3 for the study.

Model Answer: Airport Security

- Question:** Will one additional lane reduce the 90th-percentile wait?
- Classification:** dynamic, stochastic, and discrete.
- Reasons:** waiting unfolds over time, times vary, and counts change at events.
- Steps 1-3:** define the target, collect timing data, and validate assumptions with staff.

Week 2 Checklist

1. Start with the decision question.
2. Classify time, randomness, and state change independently.
3. Give a reason for every classification label.
4. Let the classification guide data, logic, and analysis.
5. Follow the study steps before trusting the result.

Next Week: Representing Uncertainty

- **Today:** stochastic models were identified by random variation.
- **Next question:** how do we describe and generate that variation correctly?
- **Next topics:** probability, random variables, distributions, and random-number generation.
- **Connection:** these tools provide uncertain inputs for stochastic simulations.

Thank you!